

JAWAD CHAUDHARY

MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN MB502 MOLECULAR BIOLOGY CONCEPT CRACKER MCQS FINAL EXAMS

Main Group (BS Biological Sciences) https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

MB502 Molecular Biology https://chat.whatsapp.com/GIcr0aNQzUv23spXjzy6AM

Q1: Matthew Meselson and Franklin Stahl labelled ___ with 15N.

Answer: E. coli DNA

Q2: A nucleosome consists of _____.

Answer: 200 bp of DNA and nine histones

Q3: A nucleosome has about ___ nucleotides in eukaryotic genome.

Answer: 200

Q4: In the cell, both strands of the DNA duplex are replicated at the ___ time.

Answer: Same

Q5: Key substrates required for the synthesis of DNA to proceed are __ and ____.

Answer: Deoxy nucleoside triphosphates and Primer

Q6: The overall three-dimensional arrangement of all atoms in a protein is referred as ___.

Answer: Tertiary structure of protein

Q7: All DNA polymerase require a primer with a free...

Answer: 3'-OH

Q8: The nucleosome of eukaryotic genome contains about __ protein molecules with DNA.

Answer: 9

Q9: The percentage of nucleic acid in relation to protein is about __% for some bacteriophages.

Answer: 50%

Q10: The 5' splice site is recognized by:

Answer: U1 SnRNP

Q11: During initial transcription, RNA polymerase produces and releases short RNA transcripts of _____.

Answer: <10 nucleotides

Q12: All eukaryotes have ____ types of RNA polymerases.

Answer: 3

Q13: During RNA splicing pre-mRNA is converted into mature mRNA containing ____.

Answer: Exons

Q14: TFIIB protein is a ____ polypeptide chain, enters the preinitiation complex.

Answer: Single

Q15: TFIIH controls the ____ transition of the preinitiation complex to the open complex.

Answer: ATP-dependent

Q16: An intron is removed through two successive ____ reactions in which phosphodiester linkages within the pre-mRNA are broken and new ones are formed.

Answer: Transesterification

Q17: The internal guide sequence of group I introns base pairs with _____.

Answer: 5' splice site sequence

Q18: A gene coding sequence is a series of ____ nucleotide codons.

Answer: 3

Q19: ___ invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.

Answer: Scrunching

Q20: Large subunit of _____ has a carboxy-terminal domain (CTD), which is referred to as the "tail".

Answer: Pol II

Q21: Find the wrong statement in the following:

Answer: The primary transcript only contains exons

Q22: RNA polymerase starts most transcripts with _____ ribonucleotide.

Answer: Adenine

Q23: TFIIE, a general transcription factor consists of ___ subunits.

Answer: 2

Q24: The coding sequence of a gene is a series of codons that specifies the ____ of amino acids in the polypeptide product.

Answer: Linear sequence

Q25: During the removal of intron, the first reaction is triggered by:

Answer: 2'-OH of the conserved A at the branch site

Q26: The branchpoint site in RNA editing is present on _____.

Answer: Introns

Q27: ____ classes of splicing found in the cell.

Answer: 2

Q28: The general transcription factors help polymerase bind to the ____ and melt the DNA.

Answer: Promoter

Q29: ___ invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.

Answer: Scrunching

Q30: Transition from initiation to elongation phase involves the _____ enzyme shedding most of its initiation factors.

Answer: Pol II

Q31: Find the wrong statement in the following:

Answer: The primary transcript only contains exons

Q32: RNA polymerase starts most transcripts with _____ ribonucleotide.

Answer: Adenine

Q33: TFIIE, a general transcription factor consists of ___ subunits.

Answer: 2

Q34: Bacteria have ___ types of RNA polymerase.

Answer: 1

Q35: TAF42 and TAF62 from Drosophila have been shown to form a structure similar to that of the _____ tetramer of histones.

Answer: H3.H4

Q36: The "transient excursion" model proposes transient cycles of forward and reverse translocation of _____.

Answer: RNA polymerase

Q37: In RNA splicing mature mRNA contains ____.

Answer: Exons

Q38: ___ invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.

Answer: Scrunching

Q39: Find the wrong statement in the following:

Answer: The primary transcript only contains exons

Q40: RNA polymerase starts most transcripts with _____ ribonucleotide.

Answer: Adenine

Q41: TFIIE, a general transcription factor consists of ___ subunits.

Answer: 2

Q42: ___ invokes a flexible element within the polymerase that allows a module at the front of the enzyme, containing the active site.

Answer: Scrunching

Q43: Find the wrong statement in the following:

Answer: The primary transcript only contains exons

Q44: RNA polymerase starts most transcripts with _____ ribonucleotide.

Answer: Adenine

Q45: TFIIE, a general transcription factor consists of ___ subunits.

Answer: 2

Q46: The “transient excursion” model proposes transient cycles of forward and reverse translocation of _____.

Answer: RNA polymerase

Q47: In RNA splicing mature mRNA contains ____.

Answer: Exons

Q48: ____ is known as the minor form of the spliceosome.

Answer: AT-AC spliceosome

Q49: During elongation, the enzyme adds _____ nucleotide at a time to the growing RNA transcript.

Answer: 1

Q50: Phosphorylation of the CTD leads to an exchange of initiation factors for those factors required for ____.

Answer: Elongation

Q51: The process of mismatch repair begins when _____ binds to the mismatched base pair.

Answer: MutS

Q52: In long-patch mismatch repair, approximately _____ nucleotides are replaced.

Answer: 2–8

Q53: The concept that genetic information flows from DNA to RNA to protein is called the _____.

Answer: Central Dogma

Q54: The phenomenon of transformation was first demonstrated in the bacterium _____.

Answer: *Streptococcus pneumoniae*

Q55: Separation (decatenation) of replicated DNA molecules is accomplished by the action of _____.

Answer: Type II topoisomerase

Q56: A nucleoside consists of a nitrogenous base and a _____.

Answer: Pentose sugar

Q57: Rosalind Franklin and Maurice Wilkins used X-ray diffraction to study the _____ structure of DNA.

Answer: Three-dimensional

Q58: Coumarin is a _____ ring compound that intercalates into the DNA molecule.

Answer: Planar

Q59: DNA damage caused by X-rays or gamma rays mainly occurs through the formation of reactive _____ from water molecules.

Answer: Free radicals

Q60: Mutation is considered a necessary evil because it produces genetic _____ required for evolution.

Answer: Variation

Q61: Mutations contribute to the occurrence of genetic _____.

Answer: Variation

Q62: The action of _____ introduces supercoils into the DNA duplex ahead of the replication fork.

Answer: DNA gyrase

Q63: The complex consisting of RNA polymerase, DNA template, and newly synthesized RNA is called the _____ complex.

Answer: Transcription

Q64: The most common mismatch base pair recognized during DNA mismatch repair in *E. coli* is _____.

Answer: G–T mismatch

Q65: A nucleoside phosphate contains _____ phosphate group(s) attached to the 5′ hydroxyl of the sugar.

Answer: One or more

Q66: UV-irradiated *E. coli* can restore their ability to survive after incubation in _____ light.

Answer: Visible

Q67: Nucleosomes fold to form the _____ fiber.

Answer: 30-nm chromatin

Q68: Rupert and co-workers isolated the photo reactivating enzyme from the Gram-negative bacterium _____.

Answer: *Escherichia coli*

Q69: Secondary structure refers to the local conformation of a portion of a _____.

Answer: Polypeptide chain

Q70: Alfred Hershey used T2 bacteriophage to infect _____ cells.

Answer: *Escherichia coli*

Q71: DNA was established as the hereditary material through a series of experiments conducted by Griffith, Avery, and _____.

Answer: Hershey and Chase

Q72: Griffith called the substance responsible for bacterial transformation the _____.

Answer: Transforming principle

Q73: The transforming principle was later identified as _____.

Answer: DNA

Q74: Avery demonstrated that bacterial transformation is prevented when DNA is treated with _____.

Answer: DNase

Q75: In the Hershey-Chase experiment, DNA was labeled with radioactive _____.

Answer: Phosphorus-32

Q76: In the Hershey-Chase experiment, proteins were labeled with radioactive _____.

Answer: Sulfur-35

Q77: The bacteriophage used by Hershey and Chase was _____ phage.

Answer: T2

Q78: The Hershey-Chase experiment proved that _____ carries hereditary information.

Answer: DNA

Q79: Watson and Crick proposed the double-helical model of DNA in _____.

Answer: 1953

Q80: DNA consists of two polynucleotide chains arranged in an _____ manner.

Answer: Antiparallel

Q81: Complementary base pairing is maintained by _____ bonds.

Answer: Hydrogen

Q82: Adenine pairs specifically with _____ in DNA.

Answer: Thymine

Q83: Guanine pairs specifically with _____ in DNA.

Answer: Cytosine

Q84: Purines pair with _____ to maintain uniform DNA diameter.

Answer: Pyrimidines

Q85: Chargaff's rule states that the amount of adenine equals the amount of _____.

Answer: Thymine

Q86: DNA replication is described as _____ because each daughter molecule contains one parental strand.

Answer: Semiconservative

Q87: DNA synthesis always proceeds in the _____ direction.

Answer: 5′→3′

Q88: The enzyme responsible for unwinding the DNA double helix is _____.

Answer: Helicase

Q89: Single-stranded DNA is stabilized during replication by _____ proteins.

Answer: SSB

Q90: DNA polymerase requires a free _____ group to begin DNA synthesis.

Answer: 3′-OH

Q91: RNA primers are synthesized by the enzyme _____.

Answer: Primase

Q92: DNA polymerase III is the primary enzyme responsible for DNA _____ in prokaryotes.

Answer: Replication

Q93: The leading strand is synthesized in a _____ manner.

Answer: Continuous

Q94: The lagging strand is synthesized _____.

Answer: Discontinuously

Q95: Short DNA fragments synthesized on the lagging strand are called _____ fragments.

Answer: Okazaki

Q96: RNA primers are removed mainly by DNA polymerase _____ in *E. coli*.

Answer: I

Q97: DNA ligase joins adjacent DNA fragments by forming _____ bonds.

Answer: Phosphodiester

Q98: DNA gyrase belongs to the family of _____ enzymes.

Answer: Topoisomerase

Q99: The action of DNA gyrase prevents excessive DNA _____ during replication.

Answer: Supercoiling

Q100: DNA replication terminates at specific DNA sequences called _____ sites.

Answer: Ter

Q101: DNA polymerase possesses _____ activity to correct replication errors.

Answer: Proofreading

Q102: Proofreading depends on the _____ exonuclease activity of DNA polymerase.

Answer: 3′→5′

Q103: The spontaneous loss of a purine base from DNA is called _____.

Answer: Depurination

Q104: Ultraviolet radiation commonly induces the formation of _____ dimers.

Answer: Thymine

Q105: Photoreactivation repairs UV-induced DNA damage using the enzyme _____.

Answer: Photolyase

Q106: UV-damaged *E. coli* recovers after exposure to _____ light.

Answer: Visible

Q107: Mismatch repair begins when the _____ protein recognizes the mismatched base pair.

Answer: MutS

Q108: In *E. coli*, the MutH protein cleaves the newly synthesized _____ DNA strand.

Answer: Unmethylated

Q109: Long-patch mismatch repair generally replaces about _____ nucleotides.

Answer: 2–8

Q110: DNA damage caused by X-rays mainly results from the formation of reactive _____ radicals.

Answer: Free

Q111: Mutations are permanent changes in the nucleotide _____.

Answer: Sequence

Q112: Mutations provide the raw material for biological _____.

Answer: Evolution

Q113: A mutation involving substitution of one nucleotide for another is called a _____ mutation.

Answer: Point

Q114: A mutation that changes one amino acid into another is called a _____ mutation.

Answer: Missense

Q115: A mutation that converts a codon into a stop codon is called a _____ mutation.

Answer: Nonsense

Q116: Addition or deletion of nucleotides not in multiples of three produces a _____ mutation.

Answer: Frameshift

Q117: Chemical mutagens that resemble normal DNA bases are known as base _____.

Answer: Analogs

Q118: Intercalating agents cause mutations primarily by inducing _____.

Answer: Frameshifts

Q119: The central dogma describes the flow of genetic information from DNA to RNA to _____.

Answer: Protein

Q120: The synthesis of RNA using a DNA template is known as _____.

Answer: Transcription

Q121: The enzyme responsible for RNA synthesis during transcription is _____.

Answer: RNA polymerase

Q122: The DNA strand used as a template for RNA synthesis is called the _____ strand.

Answer: Template

Q123: RNA is synthesized in the _____ direction.

Answer: 5′→3′

Q124: Transcription begins when RNA polymerase recognizes the _____ region.

Answer: Promoter

Q125: The DNA sequence that signals the end of transcription is called the _____.

Answer: Terminator

Q126: The first stage of transcription is called transcription _____.

Answer: Initiation

Q127: During transcription elongation, RNA polymerase moves along the DNA _____ strand.

Answer: Template

Q128: The sigma (σ) factor helps RNA polymerase recognize the _____.

Answer: Promoter

Q129: In prokaryotes, the sigma factor dissociates after transcription _____.

Answer: Initiation

Q130: Messenger RNA carries genetic information from DNA to the _____.

Answer: Ribosome

Q131: Ribosomal RNA forms the structural and catalytic part of the _____.

Answer: Ribosome

Q132: Transfer RNA functions as an adaptor during protein _____.

Answer: Synthesis

Q133: The three-nucleotide sequence on tRNA is called the _____.

Answer: Anticodon

Q134: The complementary three-base sequence on mRNA is called the _____.

Answer: Codon

Q135: The genetic code is described as triplet because each codon contains _____ nucleotides.

Answer: Three

Q136: The genetic code is considered degenerate because several codons may specify the same _____.

Answer: Amino acid

Q137: A codon specifies only one _____.

Answer: Amino acid

Q138: The initiation codon for protein synthesis is _____.

Answer: AUG

Q139: AUG codes for the amino acid _____.

Answer: Methionine

Q140: UAA, UAG, and UGA are _____ codons.

Answer: Stop

Q141: The genetic code is nearly _____ among all living organisms.

Answer: Universal

Q142: Amino acids are attached to their respective tRNAs by _____ synthetases.

Answer: Aminoacyl-tRNA

Q143: Charging of tRNA requires energy from _____.

Answer: ATP

Q144: The functional unit consisting of mRNA, ribosome, and tRNA during protein synthesis is called the translation _____.

Answer: Complex

Q145: Translation begins with the binding of the small ribosomal subunit to the _____.

Answer: mRNA

Q146: The initiator tRNA first occupies the _____ site of the ribosome.

Answer: P

Q147: The incoming aminoacyl-tRNA binds to the _____ site of the ribosome.

Answer: A

Q148: Peptide bond formation is catalyzed by the enzyme _____.

Answer: Peptidyl transferase

Q149: After peptide bond formation, the empty tRNA leaves through the _____ site.

Answer: E

Q150: Movement of the ribosome along mRNA is known as _____.

Answer: Translocation

Q151: Protein synthesis ends when a _____ codon enters the A site.

Answer: Stop

Q152: Stop codons are recognized by _____ factors.

Answer: Release

Q153: The first amino acid of a newly synthesized polypeptide is usually _____.

Answer: Methionine

Q154: In prokaryotes, the initiating amino acid is _____ methionine.

Answer: Formyl

Q155: Several ribosomes translating a single mRNA simultaneously form a _____.

Answer: Polysome

Q156: Removal of introns from pre-mRNA is called RNA _____.

Answer: Splicing

Q157: Coding regions of eukaryotic genes are called _____.

Answer: Exons

Q158: Non-coding regions removed during RNA processing are called _____.

Answer: Introns

Q159: Addition of a modified guanine nucleotide at the 5′ end forms the _____ cap.

Answer: 7-methylguanosine

Q160: The 3′ end of eukaryotic mRNA receives a _____ tail.

Answer: Poly(A)

Q161: RNA splicing occurs inside the _____.

Answer: Nucleus

Q162: Mature mRNA is transported from the nucleus to the _____.

Answer: Cytoplasm

Q163: Regulation of gene expression ensures proteins are synthesized only when _____.

Answer: Needed

Q164: A group of genes controlled by a single promoter is called an _____.

Answer: Operon

Q165: The operon model was proposed by _____ and Monod.

Answer: Jacob

Q166: The lac operon is an example of an _____ operon.

Answer: Inducible

Q167: The trp operon is an example of a _____ operon.

Answer: Repressible

Q168: Lactose acts as an _____ of the lac operon.

Answer: Inducer

Q169: In the absence of lactose, the lac repressor binds to the _____.

Answer: Operator

Q170: The operator is a DNA sequence that regulates transcription by binding a _____.

Answer: Repressor

Q171: The regulator gene of the lac operon produces the _____ protein.

Answer: Repressor

Q172: RNA polymerase binds to the _____ region to initiate transcription.

Answer: Promoter

Q173: The DNA sequence where the repressor binds is called the _____.

Answer: Operator

Q174: In the presence of lactose, the lac repressor becomes _____.

Answer: Inactive

Q175: The lac operon is activated only when glucose concentration is _____.

Answer: Low

Q176: Cyclic AMP binds with _____ protein to activate the lac operon.

Answer: CAP

Q177: Catabolite repression occurs when _____ is present in high concentration.

Answer: Glucose

Q178: The end product of the trp operon acts as a _____.

Answer: Corepressor

Q179: In the presence of tryptophan, transcription of the trp operon is _____.

Answer: Repressed

Q180: The trp operon is mainly involved in the biosynthesis of _____.

Answer: Tryptophan

Q181: Restriction enzymes recognize specific _____ DNA sequences.

Answer: Palindromic

Q182: Restriction endonucleases are naturally produced by _____.

Answer: Bacteria

Q183: Restriction enzymes protect bacteria from invading _____ DNA.

Answer: Viral

Q184: EcoRI recognizes a _____ nucleotide sequence.

Answer: Six

Q185: EcoRI produces _____ ends after DNA cleavage.

Answer: Sticky

Q186: Restriction enzymes that cut within their recognition site are called Type _____ enzymes.

Answer: II

Q187: DNA fragments with complementary overhangs are joined more easily because of _____ base pairing.

Answer: Complementary

Q188: The enzyme that joins DNA fragments is _____ ligase.

Answer: DNA

Q189: DNA ligase forms _____ bonds between adjacent nucleotides.

Answer: Phosphodiester

Q190: Recombinant DNA is produced by joining DNA from _____ sources.

Answer: Different

Q191: A cloning vector carries foreign DNA into a _____ cell.

Answer: Host

Q192: Plasmids are commonly used as cloning _____.

Answer: Vectors

Q193: Plasmids are small, circular, _____ DNA molecules.

Answer: Double-stranded

Q194: The origin of replication on a plasmid is abbreviated as _____.

Answer: ori

Q195: Antibiotic resistance genes on plasmids are used as _____ markers.

Answer: Selectable

Q196: A recombinant plasmid is introduced into bacteria by _____.

Answer: Transformation

Q197: The uptake of naked DNA by bacterial cells is called _____.

Answer: Transformation

Q198: Bacterial cells capable of taking up DNA are called _____ cells.

Answer: Competent

Q199: Heat shock temporarily increases the permeability of the bacterial _____.

Answer: Membrane

Q200: DNA amplification by PCR requires a pair of synthetic _____.

Answer: Primers

Q201: PCR stands for Polymerase Chain _____.

Answer: Reaction

Q202: PCR consists of three basic steps: denaturation, annealing, and _____.

Answer: Extension

Q203: DNA strands separate during PCR at the _____ step.

Answer: Denaturation

Q204: Primers bind to template DNA during the _____ step.

Answer: Annealing

Q205: DNA polymerase synthesizes new DNA during the _____ step.

Answer: Extension

Q206: The thermostable DNA polymerase used in PCR is _____ polymerase.

Answer: Taq

Q207: Taq polymerase is isolated from the bacterium _____.

Answer: *Thermus aquaticus*

Q208: Each PCR cycle approximately _____ the amount of target DNA.

Answer: Doubles

Q209: PCR can amplify DNA from extremely _____ samples.

Answer: Small

Q210: Gel electrophoresis separates DNA fragments according to their _____.

Answer: Size

Q211: DNA migrates toward the _____ electrode during electrophoresis.

Answer: Positive

Q212: Agarose gel is commonly used to separate _____ fragments.

Answer: DNA

Q213: Smaller DNA fragments migrate _____ than larger fragments.

Answer: Faster

Q214: Ethidium bromide is commonly used to _____ DNA bands.

Answer: Visualize

Q215: DNA bands stained with ethidium bromide are observed under _____ light.

Answer: Ultraviolet

Q216: Southern blotting is used for the detection of specific _____ sequences.

Answer: DNA

Q217: Northern blotting is used to analyze _____ molecules.

Answer: RNA

Q218: Western blotting is used for the detection of _____.

Answer: Proteins

Q219: DNA probes hybridize with _____ nucleotide sequences.

Answer: Complementary

Q220: Molecular cloning allows production of multiple copies of a specific DNA _____.

Answer: Fragment

Q221: Determining the exact order of nucleotides in DNA is called DNA _____.

Answer: Sequencing

Q222: The chain termination method of DNA sequencing was developed by _____.

Answer: Sanger

Q223: Dideoxynucleotides terminate DNA synthesis because they lack the _____ hydroxyl group.

Answer: 3′

Q224: DNA sequencing is commonly used to detect gene _____.

Answer: Mutations

Q225: The Human Genome Project aimed to determine the complete nucleotide _____ of human DNA.

Answer: Sequence

Q226: The Human Genome Project officially began in _____.

Answer: 1990

Q227: The Human Genome Project was completed in the year _____.

Answer: 2003

Q228: The human genome contains approximately _____ billion base pairs.

Answer: 3

Q229: Less than _____ percent of the human genome encodes proteins.

Answer: 2

Q230: The study of the complete set of genes is called _____.

Answer: Genomics

Q231: The study of the complete set of proteins produced by an organism is known as _____.

Answer: Proteomics

Q232: Bioinformatics combines biology with computer _____.

Answer: Science

Q233: Bioinformatics is widely used for storing and analyzing biological _____.

Answer: Data

Q234: Sequence alignment is used to identify similarities between DNA or protein _____.

Answer: Sequences

Q235: Genetic engineering involves deliberate modification of an organism's _____.

Answer: Genome

Q236: An organism carrying a foreign gene is called a _____ organism.

Answer: Transgenic

Q237: A transgene is introduced into the host genome using an appropriate _____.

Answer: Vector

Q238: Recombinant DNA technology allows the production of human _____ in bacteria.

Answer: Insulin

Q239: Human insulin produced by recombinant DNA technology is called _____ insulin.

Answer: Recombinant

Q240: Gene therapy aims to replace or repair defective _____.

Answer: Genes

Q241: In gene therapy, a functional gene is introduced into the patient's _____.

Answer: Cells

Q242: Viruses commonly used for gene delivery are called gene _____.

Answer: Vectors

Q243: Gene therapy may be performed on somatic or _____ cells.

Answer: Germ

Q244: Somatic gene therapy affects only the treated _____.

Answer: Individual

Q245: DNA fingerprinting is based on variations in _____ sequences.

Answer: DNA

Q246: DNA fingerprinting is extensively used in forensic _____.

Answer: Science

Q247: Restriction Fragment Length Polymorphism is abbreviated as _____.

Answer: RFLP

Q248: Variable Number Tandem Repeats are abbreviated as _____.

Answer: VNTR

Q249: Polymerase Chain Reaction is frequently used before DNA _____.

Answer: Fingerprinting

Q250: Molecular markers help identify genetic _____ among individuals.

Answer: Differences

Q251: Monoclonal antibodies are produced from a single clone of _____ cells.

Answer: B

Q252: Hybridoma technology is used for producing _____ antibodies.

Answer: Monoclonal

Q253: Biotechnology applies living organisms to produce useful _____.

Answer: Products

Q254: Industrial biotechnology utilizes microorganisms for large-scale _____.

Answer: Production

Q255: Fermentation is commonly used to produce antibiotics and _____.

Answer: Enzymes

Q256: Genetically modified crops are developed to improve yield and _____ resistance.

Answer: Pest

Q257: Bt cotton contains a gene obtained from _____.

Answer: *Bacillus thuringiensis*

Q258: The Bt gene produces a protein toxic to specific _____.

Answer: Insects

Q259: Golden Rice was developed to produce increased amounts of _____.

Answer: Beta-carotene

Q260: Molecular diagnostics are based on detecting specific DNA or _____ sequences.

Answer: RNA

Q261: Restriction enzymes are often called molecular _____.

Answer: Scissors

Q262: DNA ligase is often referred to as molecular _____.

Answer: Glue

Q263: A plasmid used in cloning should possess an origin of _____.

Answer: Replication

Q264: Selectable markers help identify cells containing the recombinant _____.

Answer: Plasmid

Q265: Expression vectors are designed for production of recombinant _____.

Answer: Proteins

Q266: The success of recombinant DNA technology depends largely on restriction enzymes and DNA _____.

Answer: Ligase

Q267: Molecular biology has greatly improved the diagnosis of inherited _____.

Answer: Diseases

Q268: Recombinant DNA technology has revolutionized modern _____.

Answer: Biotechnology

Q269: Advances in molecular biology have contributed significantly to personalized _____.

Answer: Medicine

Q270: The primary objective of molecular biology is to understand the molecular basis of _____.

Answer: Life

Q271: The branch of biology that studies genes at the molecular level is called _____.

Answer: Molecular biology

Q272: A recombinant DNA molecule contains DNA from _____ sources.

Answer: Different

Q273: The process of inserting foreign DNA into a vector is called DNA _____.

Answer: Cloning

Q274: Cosmids are hybrid vectors that combine plasmid and _____ DNA.

Answer: Phage

Q275: Bacteriophages are viruses that infect _____.

Answer: Bacteria

Q276: The λ (lambda) phage is commonly used as a cloning _____.

Answer: Vector

Q277: Shuttle vectors are capable of replication in _____ different host organisms.

Answer: Two

Q278: Expression vectors are specifically designed to produce recombinant _____.

Answer: Proteins

Q279: Restriction enzymes recognize DNA sequences that are usually _____.

Answer: Palindromic

Q280: Sticky ends facilitate DNA ligation because they contain complementary _____ sequences.

Answer: Single-stranded

Q281: Blunt-ended DNA fragments do not possess single-stranded _____.

Answer: Overhangs

Q282: Complementary sticky ends anneal through _____ bonding.

Answer: Hydrogen

Q283: DNA ligase requires _____ to seal phosphodiester bonds.

Answer: ATP

Q284: Transformation efficiency depends largely on the competency of bacterial _____.

Answer: Cells

Q285: Recombinant bacterial colonies are commonly identified using _____ resistance markers.

Answer: Antibiotic

Q286: Blue-white screening commonly utilizes the _____ gene.

Answer: lacZ

Q287: White colonies usually indicate successful insertion of foreign _____.

Answer: DNA

Q288: Restriction mapping determines the relative positions of restriction _____.

Answer: Sites

Q289: Southern blotting involves transferring DNA fragments onto a _____ membrane.

Answer: Nitrocellulose

Q290: DNA hybridization depends on complementary base _____.

Answer: Pairing

Q291: DNA probes are usually labeled with radioactive or _____ markers.

Answer: Fluorescent

Q292: Hybridization occurs only between _____ nucleotide sequences.

Answer: Complementary

Q293: PCR amplification requires repeated cycles of heating and _____.

Answer: Cooling

Q294: During PCR, primers determine the specificity of DNA _____.

Answer: Amplification

Q295: Real-time PCR allows monitoring of DNA amplification in _____ time.

Answer: Real

Q296: Reverse transcriptase synthesizes DNA using _____ as a template.

Answer: RNA

Q297: Complementary DNA synthesized from mRNA is called _____.

Answer: cDNA

Q298: cDNA libraries contain DNA synthesized from cellular _____.

Answer: mRNA

Q299: Genomic libraries contain fragments representing the entire organism's _____.

Answer: Genome

Q300: Recombinant DNA technology has become an essential tool in modern _____ research.

Answer: Biomedical

🚨 Concept Cracker MCQS Most Important Repeated Conceptual 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

BS Biological Sciences Team 🧬🎓

JAWAD CHAUDHARY